

Program : Diploma in Information Technology	
Course Code : 3261	Course Title: Fundamentals of Network Administration
Semester : 3	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To provide basic concepts of Computer Networks from a Network Administrator's point-of-view.

Course Prerequisites:

Topic	Course Code	Course Title	Semester
Basic Knowledge of IT skills		Introduction to IT systems Lab	1

Course Outcomes

On completion of the course students will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Outline the basic concepts of Computer Network and its Physical Layer.	15	Understanding
CO2	Illustrate basic concepts of Local Area Networks, IP Addressing and Subnetting.	14	Understanding
CO3	Summarize basic concepts of Routing Algorithms and Routing Protocols.	15	Understanding
CO4	Illustrate basic concepts of Transport layer protocols, Application layer protocols and Network Security.	14	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Outline the basic concepts of Computer Networks and its Physical Layer.		
M1.01	Outline Computer Network, Basic elements of a computer network and Network Topologies	2	Understanding
M1.02	Illustrate Transmission Mediums.	2	Understanding
M1.03	Outline Types of Networks	2	Understanding
M1.04	Summarize Switched Networks	2	Understanding
M1.05	Illustrate the duties of each layer in OSI Model	3	Understanding
M1.06	Outline the duties of each layer in TCP/IP Model	2	Understanding
M1.07	Summarize Network Devices.	2	Understanding
Contents: Computer Network – Basic Elements – Network Topologies (Bus, Star, Ring and Mesh) – Transmission Medium (Twisted Pair, Co-axial, Fiber Optic and Electromagnetic Waves) – Types of Networks (LAN, MAN, WAN, PAN and Wireless) – Switched Networks (Circuit Switched, Packet Switched) Protocol Stack – OSI Model – Functions of each layer – TCP/IP Model – Functions of Each Layer – Difference between OSI and TCP/IP models Networking Devices (NIC, Hub, Modem, Bridge, Switch, Router, Gateway and Firewall)			
CO2	Illustrate basic concepts of Local Area Networks, IP Addressing and Subnetting.		
M2.01	Summarize Medium Access Control Techniques	1	Understanding
M2.02	Summarize Ethernet and its types	1	Understanding
M2.03	Outline ARP and RARP	1	Understanding
M2.04	Summarize IP Address and IP	1	Understanding

	Addressing		
M2.05	Outline Classful IP Addressing and IP Address Classes.	1	Understanding
M2.06	Summarize IPv4 Addressing Types	1	Understanding
M2.07	Summarize IPv4 Reserved Addresses	1	Understanding
M2.08	Illustrate Packet Flow in an IPv4 Network	1	Understanding
M2.09	Summarize IPv4 Fragmentation	1	Understanding
M2.10	Outline Limitations of Classful Addressing	1	Understanding
M2.11	Illustrate Subnet Mask and Subnetting	2	Understanding
M2.12	Summarize Supernetting and Classless Inter-Domain Routing	1	Understanding
M2.13	Outline Variable Length Subnet Mask	1	Understanding
	Series Test – I	1	

Contents:

Local Area Network – Types of Network Links – Medium Access Control Techniques – Random Access Techniques (ALOHA, CSMA, CSMA/CD, CSMA/CA) – Ethernet – Types of Ethernet

ARP and RARP – Address Resolution Protocol – Reverse Address Resolution Protocol

IP Addressing – IP Address – IPv4 – IPv6 – Classful Addressing – IPv4 Address Classes (Class A, B, C, D and E) – IPv4 Addressing Types (Unicast, Broadcast and Multicast) – IPv4 Reserved Address (Loopback, Zero and Private Addresses) – Packet Flow in IPv4 Network – IPv4 Fragmentation – Limitations of Classful Addressing – Subnet Mask – Subnetting – Supernetting – Classless Inter-Domain Routing – Variable Length Subnet Mask

CO3	Summarize basic concept of Routing Algorithms and Routing Protocols		
M3.01	Illustrate Network Routing and Routing Algorithms	2	Understanding
M3.02	Summarize Classification of Routing Algorithms	2	Understanding
M3.03	Outline Routing Algorithm Metrics	2	Understanding
M3.04	Summarize Internet Architecture	1	Understanding
M3.05	Outline Routing Protocols and its classifications.	2	Understanding
M3.06	Summarize Routing Information Protocol	2	Understanding
M3.07	Illustrate Open Shortest Path First Routing Protocol	2	Understanding

M3.08	Summarize Border Gateway Protocol	2	Understanding
Contents: IP Routing – Routing Algorithm – Classification (Static vs Dynamic, Single-Path vs Multi-Path, Intra-Domain vs Inter-Domain, Link-State vs Distance Vector, Host Intelligent vs Router Intelligent) – Routing Algorithm Metrics (Path Length, Delay, Bandwidth, Load, Communication Cost and Reliability) – Internet Architecture Routing Protocols – Classification – Routing Information Protocol – Open Shortest Path First – Border Gateway Protocol.			
CO4	Illustrate basic concepts of Transport layer protocols, Application layer protocols and Network Security.		
M4.01	Illustrate Ports and Sockets	1	Understanding
M4.02	Outline Transmission Control Protocol and User Datagram Protocols	2	Understanding
M4.03	Summarize Hypertext Transfer Protocol	2	Understanding
M4.04	Outline Simple Mail Transfer Control Protocol	1	Understanding
M4.05	Illustrate File Transfer Protocol	1	Understanding
M4.06	Summarize Domain Name Server	1	Understanding
M4.07	Outline Dynamic Host Configuration Protocol	1	Understanding
M4.08	Summarize Network Security Needs, History and Four Pillars	2	Understanding
M4.09	Illustrate Network Security Terms and Security Components	1	Understanding
M4.10	Summarize Types of Attacks and Known Security Attacks	1	Understanding
M4.11	Outline Firewalls	1	Understanding
	Series Test – II	1	
Contents: TCP and UDP – Ports – Sockets – User Datagram Protocols – Transmission Control Protocol – Compare TCP and UDP Application Layer Protocols – Hypertext Transfer Protocol – Simple Mail Transfer Control Protocol – File Transfer Protocol – Domain Name Server – Dynamic Host Configuration Protocol Network Security – Needs – History – Four Pillars – Security Terms – Security Components – Types of Attacks – Known Security Attacks – Firewalls			

Text / Reference:

T/R	Book Title / Author
T1	Narasimha Karumanchi, Dr. A. Damodaram and Dr. M. Sreenivasa Rao. <i>Elements of Computer Networking: An Integrated Approach</i> . CareerMonk Publications, 2014.
R1	Todd Lammle, <i>CCNA Routing and Switching Complete Study Guide</i> , 2 nd Edition, Sybex (Wiley) Publication, 2016

Online Resources:

Sl.No.	Website Link
1	https://www.tutorialspoint.com/data_communication_computer_network/index.h tm
2	https://www.javatpoint.com/computer-network-tutorial
3	https://www.geeksforgeeks.org/computer-network-tutorials/